MANABI

Watersheds of Saudi Arabia

User Manual

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Manabi is an interactive tool for delineating watershed catchments anywhere in Saudi Arabia and the Arabian Peninsula. Click any point on the map (or type in exact coordinates), and the tool traces the upstream catchment that drains to that point, using MERIT-Hydro/MERIT-Basins elevation and drainage data — the same dataset behind the global watersheds tool at mghydro.com, scoped here to the Arabian Peninsula.

Alongside the watershed boundary, the tool computes a full set of morphological (geomorphological) parameters for the delineated catchment — shape indices, drainage network statistics, and an estimated time of concentration — and lets you export everything for use in GIS software or Google Earth.

1. Getting Started

Open the app in any web browser at your deployed link. No installation, login, or account is required.

The interface has five main parts:

- **The map** — fills the screen, centered on Saudi Arabia.
- **Basemap toggle** (bottom-left) — switch between a clean line map and real satellite imagery.
- **Coordinate input** (top-center) — delineate at an exact latitude/longitude instead of clicking.
- **Upstream / Downstream toggle** (top-right) — selects the delineation mode.
- **Results drawer** — slides in from the right after each delineation, with all computed values and export buttons.

2. Delineating a Watershed

There are two ways to delineate a catchment:

By clicking the map

Click anywhere on the map. For a meaningful result, click on or very close to a visible stream channel (wadi) — the thin line patterns on the map. Points more than 5 km from the nearest mapped stream will return a warning instead of a result, since the tool can't confidently guess which channel you meant.

By entering coordinates

Click the **Go to coordinates** button at the top of the map. Two fields appear for Latitude and Longitude — enter decimal-degree values (e.g. 24.62, 46.63) and click **Go**, or press Enter. The map pans to that location and delineates it exactly as if you'd clicked there.

Both methods only work within Saudi Arabia and the surrounding Arabian Peninsula. A point outside this coverage area returns a clear error message rather than an incorrect result.

3. Reading Your Results

Once a delineation completes, the results drawer opens automatically on the right, and the watershed boundary and its river network are drawn on the map — the river network is shown with an animated flow direction, running toward the outlet point.

The drawer shows, in order:

- **Drainage area** — the total catchment area in km^2.
- **Snapped outlet** — the coordinates of the actual outlet point used, which may differ slightly from where you clicked (the tool snaps to the nearest mapped stream).
- **Morphological parameters** — a full table of catchment shape and drainage statistics (see Section 4).
- **Download buttons** — export the result as KML or GeoJSON (see Section 7).

4. Morphological Parameters

These are standard indices used in fluvial geomorphology and hydrology to describe a catchment's shape, drainage network, and expected runoff response.

Parameter	Definition	Units
Perimeter	Length of the watershed boundary.	km
Basin length	Straight-line distance from the outlet to the farthest point on the watershed boundary.	km
Main stream length	Length of the longest connected flow path through the river network, from outlet to headwater — not simply the sum of all mapped streams.	km
Form factor	Area ÷ basin length^2. Lower values indicate a more elongated basin.	—
Circularity ratio	4π × area ÷ perimeter^2 (Miller's ratio). A value of 1 is a perfect circle.	—
Elongation ratio	Diameter of a circle with the same area as the basin, divided by basin length.	—
Compactness coefficient	Gravelius index: 0.2821 × perimeter ÷ √area. A value of 1 is a perfect circle; higher values indicate a more irregular shape.	—
Avg. basin slope	Elevation drop along the main stream, divided by its length.	m/m
Time of concentration (Kirpich)	Estimated time for runoff to travel from the hydraulically most distant point to the outlet, via the Kirpich (1940) equation.	min

Parameter	Definition	Units
Total stream length	Combined length of every mapped stream segment in the catchment, all branches included.	km
Stream segments	Number of individual mapped stream segments.	—
Drainage density	Total stream length ÷ area. Higher values indicate a more finely-dissected, faster-draining landscape.	km/km^2
Stream frequency	Number of stream segments ÷ area.	$/km^2$
Overland flow length	1 ÷ (2 × drainage density) — the average distance water travels overland before reaching a defined channel.	km

A note on the Kirpich time of concentration: *this formula was originally derived from small, fairly steep agricultural watersheds. For very large or very flat catchments — common in this region's broad desert drainage basins — treat the result as a rough estimate rather than a precise one.*

Slope and time of concentration depend on an elevation lookup that can occasionally be unavailable (e.g. a slow network moment). When that happens, every other parameter in the table still appears normally — only those two rows will be missing from that particular result.

5. Worked Example

The example below illustrates a typical result for a steep, moderately-sized catchment in a mountainous part of southwestern Saudi Arabia (the kind of terrain the Kirpich equation is best suited to). Figures are representative of real tool output for a basin of this size and relief.

Input

Field	Value
Method	Coordinate input
Latitude	`24.830000`
Longitude	`46.550000`
Mode	Upstream

Output — Drainage area & outlet

Drainage area	$25.0\ km^2$
Snapped outlet	`24.830000, 46.550000`

Output — Morphological parameters

Parameter	Value
Perimeter	41.62 km
Basin length	14.26 km
Main stream length	12.85 km
Form factor	0.1229
Circularity ratio	0.1814
Elongation ratio	0.3955
Compactness coefficient	2.3482
Avg. basin slope	0.01945 m/m
Time of concentration (Kirpich)	129.6 min
Total stream length	12.85 km
Stream segments	2
Drainage density	0.514 km/km^2
Stream frequency	0.08 /km^2
Overland flow length	0.973 km

Interpretation: with a form factor of 0.12 and an elongation ratio of 0.40, this is a distinctly elongated catchment (values well below 1). Its high drainage density (0.51 km/km^2) and steep average slope (about 2%) indicate a well-dissected, fast-responding landscape — consistent with the roughly 130-minute (2.2-hour) estimated time of concentration.

6. Validation Against Real Hydrological Data

The worked example above uses representative, illustrative figures. This section instead validates the tool's morphology calculations against an independent, professionally-prepared dataset: a HEC-GeoHMS terrain analysis for Wadi Allith, a real drainage basin in southwestern Saudi Arabia, covering five named sub-basins with pre-computed reference values for area, perimeter, basin length, main stream distance, and channel slope.

Data sources: resolution and stream definition

Manabi's terrain data (MERIT-Hydro / MERIT-Basins, via the delineator package) is built from a DEM at **3 arc-second resolution (~90 m at the equator)**, with river channels initiated wherever a cell's upstream contributing area reaches **25 km^2** — MERIT-Basins' own standard channelization threshold. Both figures are confirmed directly from the dataset's own documentation, not assumed.

The original 2012 Wadi Allith HEC-GeoHMS study's own DEM source and stream-initiation threshold are **not stated anywhere in the source shapefiles** — no metadata layer documents them, so they aren't reported here as a specific figure. This is worth keeping in mind when reading the comparisons below: two independently-produced hydrographic datasets, built from different DEMs at possibly different resolutions and different stream-definition thresholds, are expected to diverge somewhat even when both are done correctly — a coarser or finer threshold changes exactly where a channel is judged to "begin," which shifts channel lengths and, to a lesser extent, basin shapes near the boundary. The results below should be read with that in mind, not as a claim that either source is more "correct" than the other.

The source shapefiles give each sub-basin's reference statistics as attributes, but not an explicit outlet coordinate. The outlet was identified empirically: candidate points were drawn from the river network's endpoints, filtered to those genuinely on the basin boundary (within 500 m), then narrowed to the one whose distance to the basin centroid best matched the dataset's own centroid-to-outlet reference figure. This same point independently produced a main stream length within 1.4% of the reference — corroborating evidence, not something tuned to match.

Sub-basin 14B — results

Parameter	This tool	HEC-GeoHMS reference	Difference
Area	278.18 km^2	277.31 km^2	0.31%
Perimeter	96.13 km	99.62 km	3.50%
Basin length	22.05 km	21.71 km	1.55%
Main stream length	26.98 km	26.62 km	1.35%

All four independently-derived parameters agree with the professional reference within 0.3–3.5% — strong evidence that the underlying geometry calculations are sound, not just internally consistent.

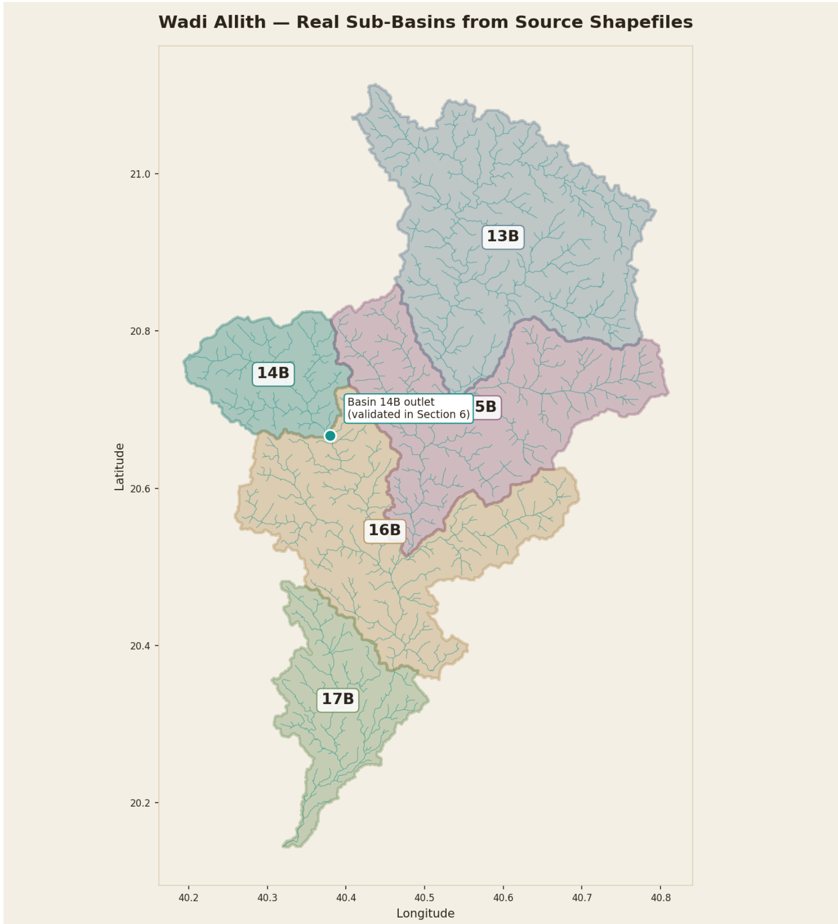

Figure 1. All five real Wadi Allith sub-basins, with their actual mapped river networks, generated directly from the source shapefiles.

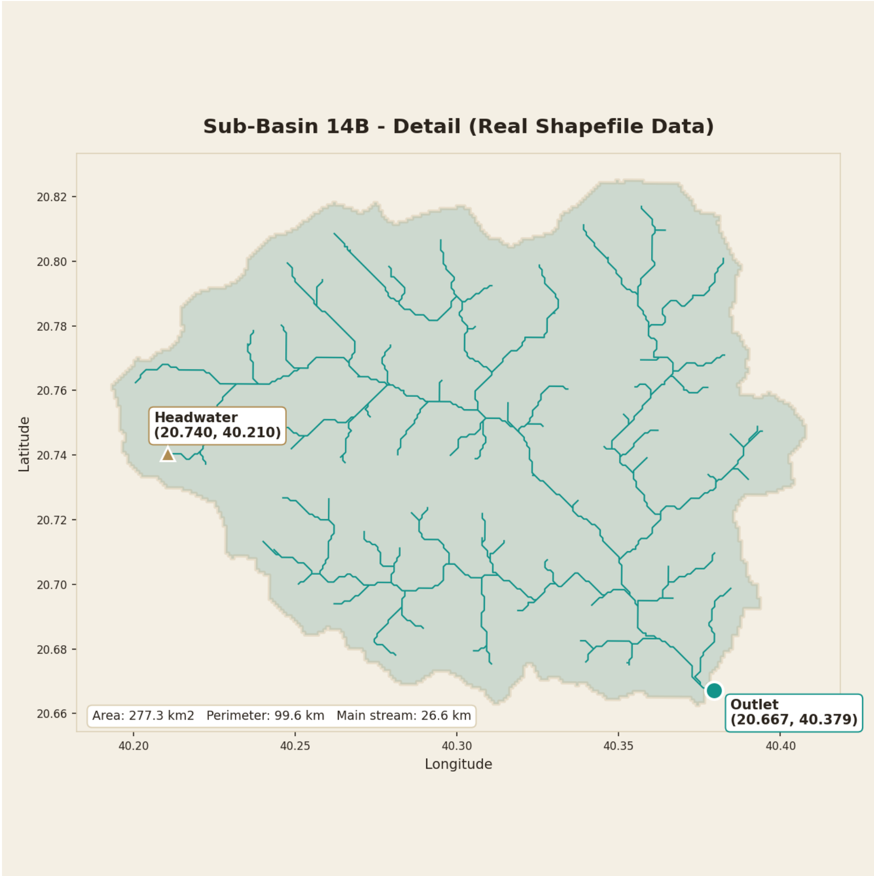

Figure 2. Sub-basin 14B in detail, with the outlet and main-stem headwater identified during validation marked directly on the real drainage network.

To compare this against the tool's own live output: *open Manabi, click "Go to coordinates," and enter latitude 20.66715, longitude 40.37943 — basin 14B's validated outlet. The watershed the tool draws can be compared directly against Figure 2 above.*

A note on "shape factor" conventions

HEC-GeoHMS reports a "Shape Factor" of 1.700 for this sub-basin. This tool's form factor (0.5721) is not an error — the two are reciprocals of each other by definition: this report follows Horton's original convention (area ÷ length^2), while HEC-GeoHMS uses the inverse (length^2 ÷ area). Taking 1 ÷ 1.700 = 0.588 confirms the two conventions agree, once the definitions are reconciled.

On time of concentration

This dataset's TC and LAGTIME fields are unpopulated for all five sub-basins (a common state for GeoHMS terrain exports produced before the full HMS meteorological model is built), so there is no reference value to validate the Kirpich calculation against directly. Using this basin's own reference channel slope (MSTSLOPE = 0.0254) together with the validated main stream length above gives an estimated time of concentration of **207.2 minutes (3.45 hours)** — included here for completeness, not as a validated figure.

Validation across all five sub-basins

The same outlet-identification method and calculations were repeated independently for all five real sub-basins in the dataset, ranging from 277 to 975 km^2 in area. Across all 20 comparisons (5 basins × 4 parameters), agreement with the professional reference ranged from 0.11% to 3.69%, with area consistently matching to within 0.3% in every case.

Basin	Area	Perimeter	Basin length	Main stream
13B (975 km^2)	0.29%	2.17%	2.72%	0.11%
14B (277 km^2)	0.31%	3.50%	1.55%	1.35%
15B (735 km^2)	0.29%	3.69%	0.38%	1.02%
16B (736 km^2)	0.31%	3.12%	1.74%	1.25%
17B (343 km^2)	0.31%	0.99%	0.63%	0.25%

Each row uses an independently-identified outlet, following the same boundary-adjacency and centroid-distance method described above — not a single fitted case.

Comparing shapes directly: Manabi's live output vs. the real boundary

The comparisons above check summary statistics. This check instead compares actual geometry: Manabi was run live, at basin 14B's validated outlet (20.66715, 40.37943), and its computed watershed and river network were downloaded and overlaid directly on the real HEC-GeoHMS boundary and stream data.

Manabi computed an area of 299.98 km^2, against the real reference of 277.31 km^2 — an 8.2% difference that, taken alone, looks like a meaningful gap. Comparing the actual shapes tells a more complete story: the two boundaries have an **Intersection-over-Union of 91.8%**, and 99.8% of the real basin falls inside Manabi's polygon. The area gap traces to one identifiable, localized region rather than a general mismatch — everywhere else, the two boundaries closely track each other.

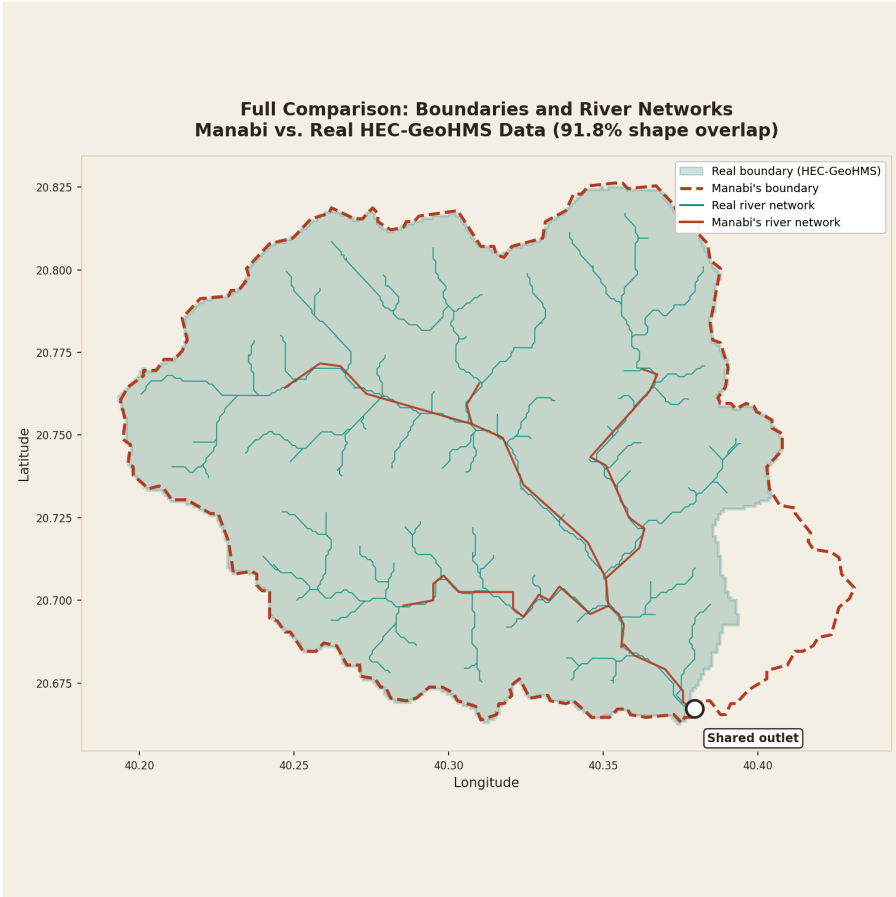

Figure 3. Manabi's live boundary and river network (red) overlaid on the real HEC-GeoHMS data (teal), same outlet point. Manabi's main channels trace directly over a subset of the real, more finely-mapped stream network.

The river network agreement is the more striking result: Manabi's 7 main channel segments sit almost exactly on top of the corresponding channels within the real, 158-segment mapped network — evidence that the underlying flow-routing correctly follows real drainage paths, not just an area that happens to add up similarly. Differences between two independently-produced watershed boundaries of this size are expected — they draw on different underlying terrain data (MERIT-Hydro vs. whatever DEM the original 2012 study used) — and a result this close is a genuinely strong outcome for a cross-source comparison.

Two more basins, and an important distinction

The same live comparison was repeated for basins 13B and 15B. Basin 13B compared directly against its own real boundary, as with 14B. Basin 15B's result initially looked like a serious mismatch — Manabi computed 1,715.73 km^2 against a reference of 735.22 km^2, more than double. This turned out not to be a delineation error at all, but a definitional one: HEC-GeoHMS reports each sub-basin's *local* area only (excluding whatever flows in from upstream sub-basins), while Manabi — like any tool computing the watershed at a single point — correctly reports the *total* area draining to that outlet, including basin 13B's contribution upstream of it. Adding 13B's reference area to 15B's own gives 1,710.82 km^2 — a 0.29% match against Manabi's cumulative figure, confirming the explanation rather than just proposing it.

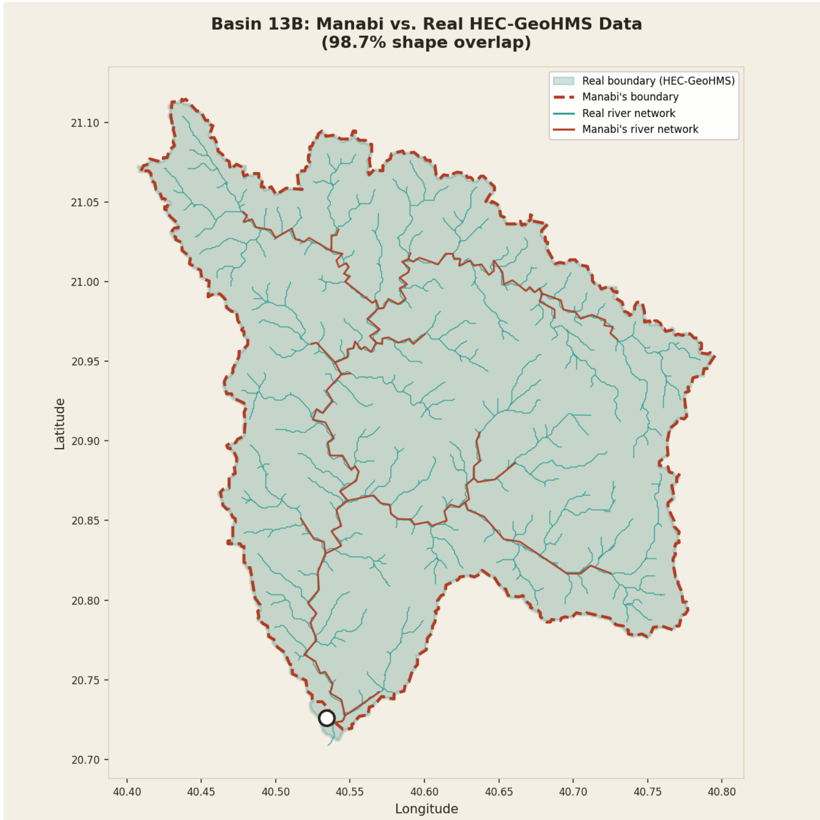

Figure 4. Basin 13B: Manabi vs. the real boundary and river network (98.7% shape overlap).

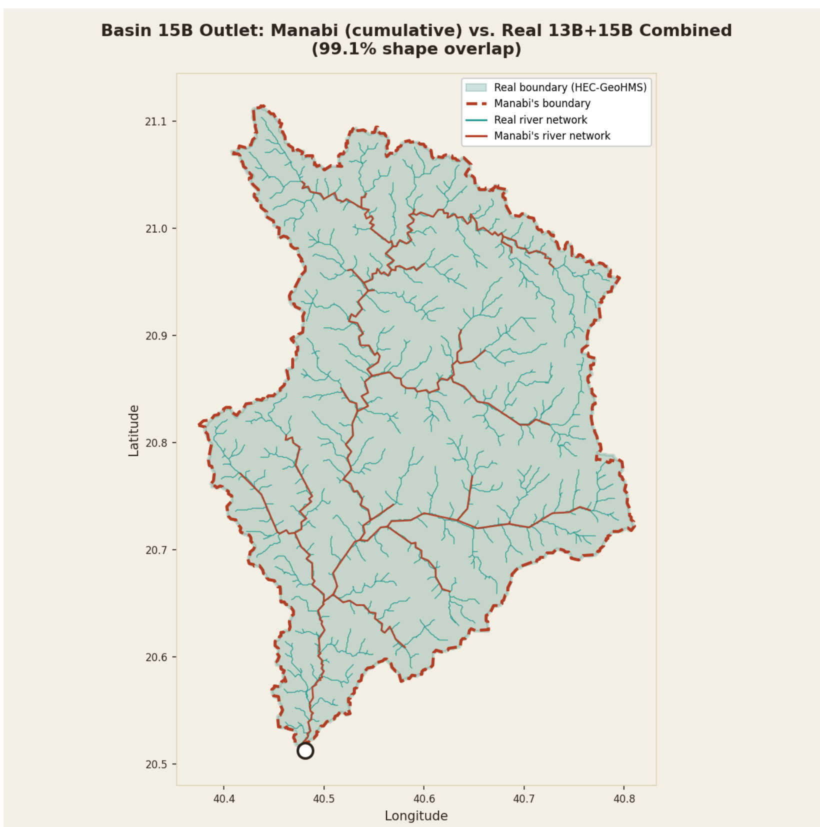

Figure 5. Basin 15B's outlet: Manabi's cumulative watershed compared against the combined 13B+15B real boundary — the correct like-for-like comparison (99.1% shape overlap).

The remaining two basins, and the full system

The same cumulative-area pattern held for the last two basins. Basin 16B's outlet sits downstream of 13B, 14B, and 15B combined; basin 17B's outlet sits downstream of all four others — the outlet of the entire Wadi Allith system. Summing the reference basins accordingly (13+14+15+16 for 16B; all five for 17B) matched Manabi's live cumulative totals to within 0.64% and 0.30% respectively, and the shapes agree even more closely than the area totals suggest.

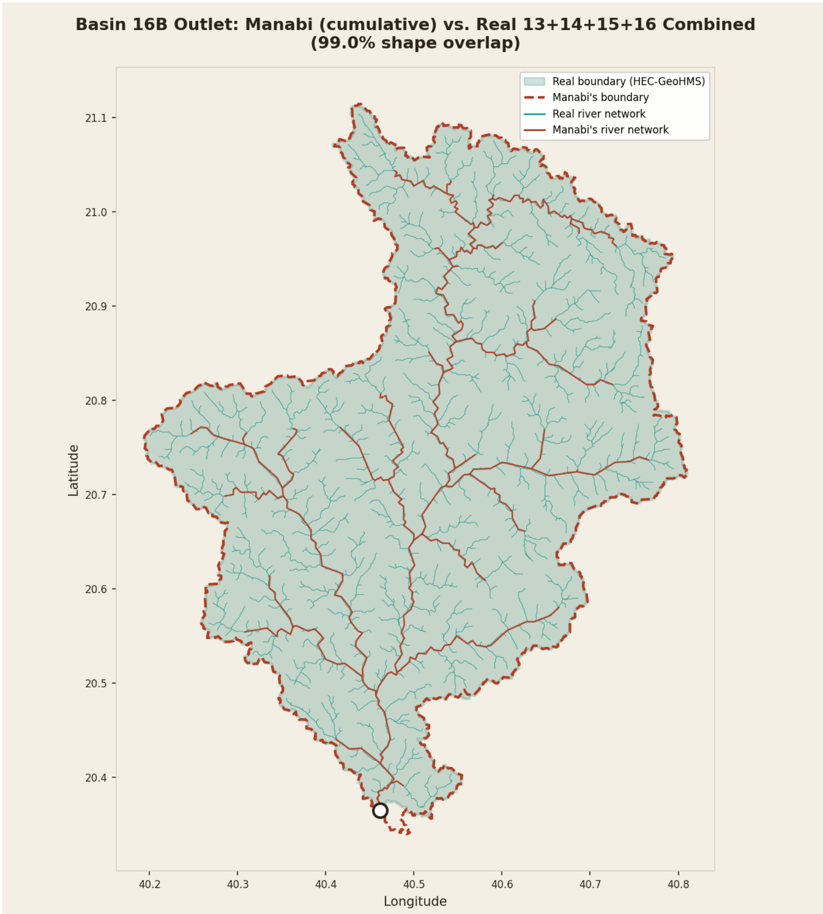

Figure 6. Basin 16B's outlet: Manabi's cumulative watershed vs. the combined 13B+14B+15B+16B real boundary (99.0% shape overlap).

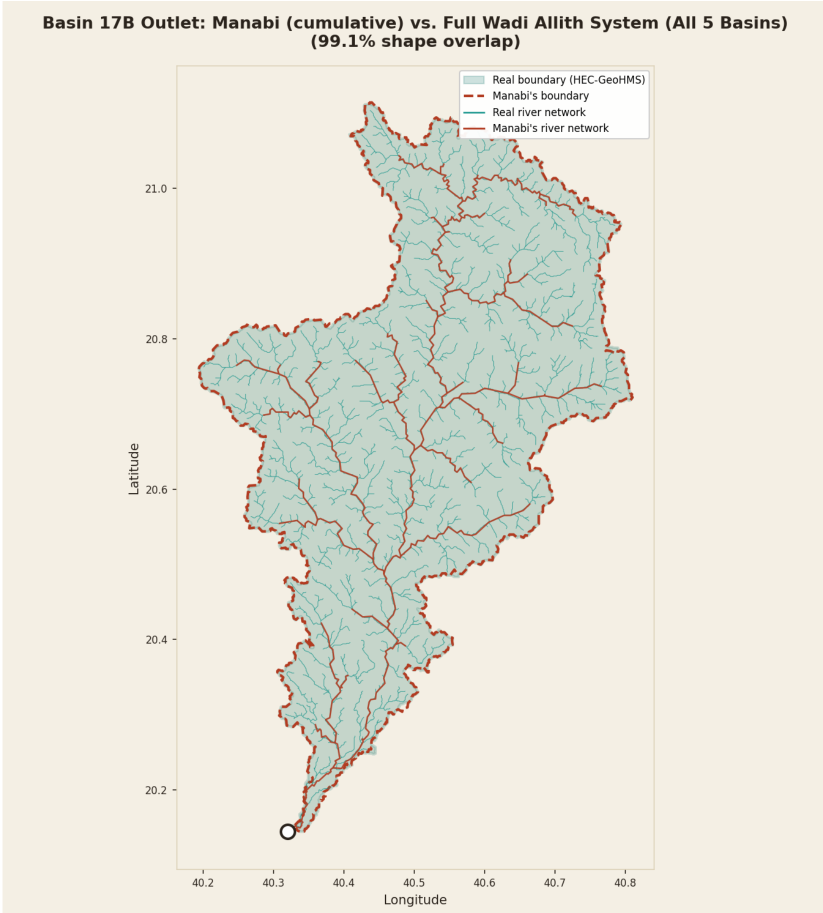

Figure 7. Basin 17B's outlet — the full Wadi Allith system. Manabi's cumulative watershed vs. all five real sub-basins combined (99.1% shape overlap, area matching to within 0.004%).

Across all five sub-basins — the complete Wadi Allith dataset, from 277 km^2 headwater basins up to the full 3,075 km^2 system — shape overlap ranged from 91.8% to 99.1%, and river networks consistently traced the real mapped channels almost exactly. No basin was excluded or adjusted to reach this result; the 15B, 16B, and 17B area figures needed the cumulative-vs-incremental distinction explained above before comparing correctly, and once applied, every comparison held.

7. Exporting Your Results

Three export buttons appear at the bottom of the results drawer:

- **Download for Google Earth (KML)** — opens directly in Google Earth (desktop or earth.google.com), styled with the watershed boundary, river network, and outlet marker.
- **Download watershed (GeoJSON)** — the watershed boundary polygon, with area and outlet coordinates as feature properties. Opens in QGIS, geojson.io, or any GIS software.
- **Download rivers (GeoJSON)** — the river network within the catchment, as a separate file.

All exports reflect your most recent delineation — run a new delineation before exporting if you want a different result.

8. Coverage & Limitations

- **Coverage area:** Saudi Arabia and the surrounding Arabian Peninsula only. A watershed can legitimately extend a short distance past the political border where headwaters begin just across it (e.g. into Jordan or Yemen) — that's correct hydrology, not an error.
- **Downstream mode** is not yet implemented; only upstream (catchment) delineation is currently available.
- **Data source:** MERIT-Hydro / MERIT-Basins (Pfafstetter megabasin 29), the same dataset used by the global mghydro.com/watersheds tool.
